Echoes in AI: Quantifying Lack of Plot Diversity in LLM Outputs

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With rapid advances in large language models (LLMs), there has been an increasing application of LLMs in creative content ideation and generation. A critical question emerges: can current LLMs provide ideas that are diverse enough to truly bolster collective creativity? We examine two state-of-the-art LLMs, GPT-4 and LLaMA-3, on story generation and discover that LLM-generated stories often consist of plot elements that are echoed across a number of generations. To quantify this phenomenon, we introduce the *Sui Generis* score, the first automatic metric that measures the uniqueness of a plot element among alternative storylines generated using the same prompt under an LLM. Evaluating on 100 short stories, we find that LLM-generated stories often contain combinations of idiosyncratic plot elements echoed frequently across generations and across different LLMs, while plots from the original human-written stories are rarely recreated or even echoed in pieces. Moreover, our human evaluation shows that the ranking of *Sui Generis* scores among story segments correlates moderately with human judgment of surprise level, even though score computation is completely automatic without relying on human judgment.

LLM | text generation | deep learning | creativity | AI

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Significance Statement

Reading through a set of texts generated by LLMs under the same prompt can be a disappointing experience. The first might seem convincingly like a novel, creative work by a human writer. Read more, though, and the lack of diversity in these LLM-generated outputs reveals itself. We show that short stories generated in this way often contain repetitive combinations of plot elements, while human-written stories maintain a higher level of uniqueness. This research is the first to quantify this observation, and to introduce an automatic metric aimed at measuring the usefulness of LLMs for creative content generation at the narrative level. We believe that this metric will help drive progress towards LLMs that generate more diverse and creative content.

1. Introduction

Rapid advances in large language models (LLMs) have spurred an ongoing debate on the usefulness of these models for tasks that require human-level creativity. On the one hand, there are works that highlight the strengths of LLMs in creative writing (1, 2), poetry generation (3), idea generation (4, 5) and even creative thinking (6). On the other hand, there has been research arguing that LLM creativity is much weaker than human creativity (7) and that LLM-generated stories are identifiably bad (8, 9).

One recent study (10) finds that while the use of an AI assistant in writing appears to enhance the creativity of individual writers, it also reduces the collective diversity of novel content produced by multiple writers. Similarly, a user study on argumentative essay writing (11) finds that writing with LLMs reduces the diversity of content produced by a group of users. These findings reflect a phenomenon that is by now familiar* to teachers whose students use LLMs for help with writing assignments: while any individual LLM output might seem compelling and novel, reading through multiple texts produced by the same prompt can be a deflating experience. A narrative element that seems strikingly innovative and creative when encountered in the first output may begin to seem, by the time it is encountered in 10 more outputs, more like the product of a deterministic process than human-like creativity.

An example of this phenomenon is shown in Table 1, which shows a small subset of 100 short story continuations generated by GPT-4† given the first part of *Give It Up*, a short story by Franz Kafka‡ (see the full list of LLM-generated continuations in supplementary material). None of GPT-4's continuations resemble Kafka's own ending for the story, in which a policeman disconcertingly tells the direction-seeking narrator to "Give it up!" and abruptly turns away. Instead, in 50 out of 100 generations, the policeman gives instructions to take the second left; in 18/100 to take the second right; and in others walks with the protagonist to show them the way. In 16/100 of the generations, a bakery is mentioned as a landmark. These *echoes*, as we refer to them here, repeat surprisingly frequently across generations at all semantic levels and not necessarily in the same order. The previously observed reduction in lexical diversity of LLM outputs (12) corresponds to what we view as semantically lower-level echoes, such as the frequent use of word "bakery." However, lexical diversity cannot be used to explain or quantify echoes at the higher semantic level. Narrative developments such as the policeman's decision to show the way or point the protagonist towards a specific direction can be paraphrased in different ways, as the example continuations show. These more abstract echoes are not merely the result of repeated tokens.

*https://edtechrce.org/educators-develop-strategies-to-detect-ai-generated-student-work/

†We use temperature $\tau = 1$ as LLMs are trained with the goal of fully capturing the data distribution with $\tau = 1$.

‡https://www.flashfictiononline.com/article/give-it-up/

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In this paper, we aim to quantify such repetitions at the *narrative* level by introducing a *Sui Generis* score.[§] We use story generation as a testbed. To compute the *Sui Generis* score for a story, we first ask the LLM to generate many alternative continuations for the same story given varying lengths of the story prefix as context. We then count the number of times an original story segment is echoed (at the narrative level) in the alternative continuations. We adopt the intuition that a segment is more likely to appear in other story samples and is thus less unique if it is echoed in a larger number of alternative continuations given less context from the previous plot.

We test the *Sui Generis* score on 100 stories consisting of around $3,700$ segments from two datasets: WritingPrompts, which contains short stories posted on Reddit; and a set of television episode plot summaries from Wikipedia. We experiment with two state-of-the-art LLMs: GPT-4 and LLaMA-3. Our results highlight the lack of diversity in LLM outputs: LLM-generated stories are often composed of the same combinations of idiosyncratic plot elements that are echoed frequently across its own generations and even across different LLMs, while only rarely exhibiting plot elements found in human narratives sparked by the same prompts. As the *Sui Generis* score is automatically calculated without human judgment, this result provides a quantitative explanation for the previous qualitative findings that the use of an AI assistant appears to reduce output collective diversity (10). It explains the contradictory findings in previous user studies on whether LLM generations have reached human-level creativity, as our study suggests that an LLM generation may seem novel to a human who has not seen many of its generations in that domain, but may seem banal to someone more familiar with its outputs. Furthermore, our human evaluation of the surprise level of story segments shows that the *Sui Generis* score correlates moderately with the average human judgment of surprisal. We observe that the story segments with high scores often correspond to key plot elements or interesting turning points, whereas those with low scores correspond to developments that are bland or highly predictable from the context.

2. Related Work

LLMs are increasingly used in creative writing (4, 13). However, there is a debate on whether LLMs boost creativity. Some studies have suggested that LLM-generated content is considered more creative or preferred by users. For example, (14) finds that ideas generated by ChatGPT were more likely to be purchased than those generated by Wharton MBA students. (4) find that when participants were asked to generate creative ideas for everyday purposes, the use of ChatGPT increased their creativity. On the other hand, (15) finds that AI-generated narratives tend to be less imaginative and only occasionally include plot twists. (7) invite expert writers to rate the stories generated by LLMs versus professional writers based on Torrance Test of Creative Writing and discover that LLM-generated stories are less creative than those of professionals.

While these works all focus on evaluating the creativity or novelty of each individual LLM output, (11) and (10) discover that, although writers report an AI assistant being helpful in their creative writing, it reduces the collective diversity of content produced by multiple writers. Similarly, (16) shows that ideas generated with the assistance of LLMs are as diverse as the ideas generated without LLMs at the individual level, but are significantly less diverse at the group level. These findings suggest that we should examine the *distribution* of LLM creations for a *given* prompt instead of each creation individually.

Prompt: It was very early in the morning, the streets clean and deserted, I was walking to the station. [...] I did not yet know my way very well in this town; luckily, a policeman was nearby, I ran up to him and breathlessly asked him the way. He smiled and said: "From me you want to know the way?" "Yes," I said, "since I cannot find it myself."
Original Continuation: "Give it up! Give it up," he said, and turned away with a sudden jerk, like people who want to be alone with their laughter.
Continuation 1: "Follow this street for two blocks, then take a left at the bakery, and you'll see the station just ahead," he instructed with a gentle nod.
Continuation 2: "Keep going straight for two blocks, then take a left and you'll see the station right ahead," he explained, giving a friendly nod.
Continuation 3: He nodded in understanding, his smile broadening as he pointed down the street, "Take the second left and cross the square; you'll see the station right ahead."
Continuation 4: "Go straight ahead and take your second left, pass the old library, and the station will be right before your eyes," he responded, tipping his cap.
Continuation 5: "Alright," he said, his smile broadening, "I'll guide you. Follow me." Grateful, I trailed behind him, my earlier panic subsiding with each step.

Table 1. Continuations generated by GPT-4 when prompted with the first part of a short story by Kafka.

A recent work on detecting novelty in LLM outputs suggests that "the text must not have been copied from the training data" (17). This definition appears too narrow and surface-level: nothing resembling the original Kafka text is found in the story continuations by GPT-4 (Table 1). Yet by this standard, the generated continuations in story must be treated as novel, even though to human readers it is the original Kafka's ending that is unusual while the generated continuations are quite conventional and lack diversity. Other existing metrics that measure diversity in LLM outputs focus either on lexical features, such as ngram overlap between output texts (18) or key-point summarizations of the texts (11), or topic-level features, such as the semantic distance between document embeddings (10). However, lexical features are insufficient for measuring diversity, as (19) observe, "identifying novel text is not straightforward because the text may have less lexical overlap yet convey the same information," while topic-level features are too high-level to assess the novelty of creative works in which novelty may reside in detailed plots. Thus, in our work, we look beyond lexical or topic-level diversity and introduce the *Sui Generis* score to measure the uniqueness of text spans at the narrative level.

3. Method

We evaluate the uniqueness of a story segment based on the "alternative continuations" generated by LLMs themselves. Formally, in a story (string) S segmented into n segments $S = (s_1, s_2, ..., s_n)$ (as shown in Figure 1(c) with square brackets), at any point $j \in [1, n-1]$, we truncate the story to its prefix $S_{-j} = (s_1, s_2, ..., s_j)$ and consider replacing suffix $S_j = (s_{j+1}, s_{j+2}, ...)$ by sampling possible alternative continuations C_j from the model K times:

$$C_j^k \sim p_{LLM}(\cdot \mid S_{-j}), \quad k \in \{1, ..., K\}. \qquad [1]$$

Each string (S_{-j}, C_j^k) is thus a full story with its first part taken from the given story S (either human-written or LLM-generated)

[§] Sui Generis is a Latin phrase meaning "of its own kind," used in English biology and law literature to indicate something unlikely to be repeated or recreated.

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Alternative Continuation C_8^1

[Prompt: In the year 1989, a police investigator arrives at the site of an impossible accident: a 2016 Toyota Corolla is wrapped around a lamp post. Title: Incident: Time-ly Collision In the quaint and ordinarily serene town of Clearwater, Nevada, headlines of an inexplicable incident stole the buzz in the]
[year 1989. Snake-hipped police investigator, Dale McQuarrie, renowned for his observational skills and analytical mind, had been summoned to decipher an extraordinary accident. A car was embraced in an implausible collision—wrapped around a lamp post, as if both were caught in a dance of destruction. But it was not just]
[the peculiar accident that captured everyone's interest. It was the car itself. The metallic, midnight-blue vehicle was labeled as a 2016 Toyota Corolla, a model and make twenty-seven years ahead of its time.]
[...][...][...]
[... For the following weeks, McQuarrie delved into the investigation, relentlessly probing every lead, examining each thread that the accident]
[offered. He contacted Toyota, where after numerous heated exchanges between baffled and skeptical executives, he was assured that the car indeed was a future model yet to be designed. Intriguingly, the man identified as Claude Matthews, a theoretical physicist from Caltech, remained in a comatose state. He was reported missing]

End of Story Prefix

from a conference on quantum mechanics and time-space anomalies just two days before the accident. McQuarrie, with a growing suspicion that this was no ordinary case, decided to visit Matthews' lab at Caltech. He was met with resistance, but his badge and the gravity of the situation eventually granted him access.
...

(a)

Alternative Continuation C_2^1

[Prompt: In the year 1989, a police investigator arrives at the site of an impossible accident: a 2016 Toyota Corolla is wrapped around a lamp post. Title: Incident: Time-ly Collision In the quaint and ordinarily serene town of Clearwater, Nevada, headlines of an inexplicable incident stole the buzz in the]
[year 1989. Snake-hipped police investigator, Dale McQuarrie, renowned for his observational skills and analytical mind, had been summoned to decipher an extraordinary accident. A car was embraced in an implausible collision—wrapped around a lamp post, as if both were caught in a dance of destruction. But it was not just]

End of Story Prefix

any ordinary car. It was a 2016 Toyota Corolla, a model that wouldn't exist for another 27 years. The vehicle's sleek design and unfamiliar technology baffled the onlookers who gathered around the scene, whispering theories and snapping polaroids. Dale approached the car with a mix of caution and curiosity. The license plate was smeared with dirt, but he could make out the registration—it was from the future, too. He circled the car, taking in every detail: the airbags deployed, the dashboard lit with strange icons, and the scent of new car mixed with the acrid tang of burnt rubber and metal.
The driver's door was ajar. **Inside, slumped over the wheel, was a man in his mid-thirties, unconscious but alive.** He wore clothes of a style not yet seen in the late '80s, his smartphone clutched tightly in one hand—a device that would seem like magic to the townsfolk of Clearwater.
...
The ambulance arrived, and the medical team carefully extracted the man from the car...
He pocketed the device, feeling its weight and the warmth it emitted. He needed answers, and he knew just the person who might help.
...

(b)

Original Story

[Prompt: In the year 1989, a police investigator arrives at the site of an impossible accident: a 2016 Toyota Corolla is wrapped around a lamp post. Title: Incident: Time-ly Collision In the quaint and ordinarily serene town of Clearwater, Nevada, headlines of an inexplicable incident stole the buzz in the]
[year 1989. Snake-hipped police investigator, Dale McQuarrie, renowned for his observational skills and analytical mind, had been summoned to decipher an extraordinary accident. A car was embraced in an implausible collision—wrapped around a lamp post, as if both were caught in a dance of destruction. But it was not just]
[...][...][...]
[... Inside the Toyota, sat an unconscious man, his hands still gripping the steering wheel, his sandy]
[hair matted with blood. McQuarrie signaled for the paramedics, who quickly cut the man free and rushed him to the hospital, leaving McQuarrie to unravel the mystery behind the untimely vehicle. For the following weeks, McQuarrie delved into the investigation, relentlessly probing every lead, examining each thread that the accident]
[offered. He contacted Toyota, where after numerous heated exchanges between baffled and skeptical executives, he was assured that the car indeed was a future model yet to be designed. Intriguingly, the man identified as Claude Matthews, a theoretical physicist from Caltech, remained in a comatose state. He was reported missing]
[a few days before the accident, last seen working late in his lab on an undisclosed project. More questions were piled onto McQuarrie's investigation, with answers seemingly as evasive as the car's origins. Driven by an odd mixture of skepticism and desperation, McQuarrie decided to confront the impossibility, visiting Caltech.]
...

(c)

Fig. 1. Two continuations of the same story with (a) longer prefix, with $j = 8$ segments, and (b) shorter prefix, with $j = 2$ segments from the same original story, partially shown in (c). Segments are delineated with []. We highlight two segments in (c), $i = 7$ in blue and $i = 9$ in red, that are echoed in these alternative continuations. The red one (visiting a scientist's lab) is echoed only in continuations conditioned on a long story prefix, while the blue one (discovering a driver inside the car) is echoed frequently even given a short prefix. The *Sui Generis* score more severely penalizes the echoes discovered given shorter prefix (indicating that similar plot is more likely to be repeated by the LLM).

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and the second part being one possible way an LLM would finish it. Figure 1 shows examples of such continuations with different prefixes: with (a) $j = 8$ and (b) $j = 2$ segments from the analyzed story in (c).

Next, to evaluate how often segment s_i is repeated across generations, we compute its *echo* scores by comparing it to the alternative continuations $\{C_1^k\}, \{C_2^k\}, ..., \{C_{i-1}^k\}$ generated from varying lengths of story prefix $S_{-1}, S_{-2}, ..., S_{-(i-1)}$. Specifically, we compute the *echo* score $p_{i,j}$, which is an estimated likelihood that similar plot in segment s_i appears in an alternative continuation:

$$p_{i,j} \approx \frac{1}{K} \sum_k a(s_i, C_j^k) \qquad [2]$$

where $a(s, C)$ is a binary function that indicates if segment s or its analog is present in continuation C. For instance, in Figure 1, both $a(s_7, C_8^1)$ and $a(s_9, C_2^1)$ should be 1. We automate this function by prompting GPT-4 using the prompt template shown in Figure 2, as our human evaluation shows that GPT-4's judgment correlates well with human judgment on this task (see Section C).[¶]

The *echo* scores $p_{i,j}$ thus signal how likely and how early in the story (indicated by position j) the plot in segment s_i is suggested to be generated. We note that the earlier j is, the less likely it is that such plot element will be repeated by the LLM in an alterative continuation, therefore echoes induced earlier should be

[¶]Note that the granularity of the function $a(s, C)$ can be easily changed to capture lower-level repetitions (e.g. the policeman often suggests to take the second left in various continuations in Table 1) by adapting the prompt template.

Here is the beginning of a story: [Story prefix S_{-j}] ***Here is a continuation of the story:*** [Alternative continuation C_j] ***Does the above continuation contain any plot segment similar to the following (disregarding details like character names, specific objects in the scene, or exact dates)?*** [Story segment s_i] ***Let's think step by step.***
[LLM's step-by-step reasoning]
Finally, give a YES or NO answer. So the answer is
[LLM's final answer (YES/NO)]

Fig. 2. Prompt template used for estimating function $a(s_i, C_j)$, i.e. if the plot in segment s_i is present in continuation C_j. The texts in black are part of the prompt while the texts in pink should be generated by the LLM.

weighted more heavily. Thus, we compute the *Sui Generis* score of segment s_i by taking the weighted average of the negative log of echoes $\{-\log p_{i,1}, -\log p_{i,2}, ..., -\log p_{i,i-1}\}$ where we give higher weights to $-\log p_{i,j}$ with smaller j's:

$$\text{SG}_i = \frac{-\sum_{j=1}^{i-1} \lambda^j \cdot \log p_{i,j}}{\sum_{j=1}^{i-1} \lambda^j} \qquad [3]$$

where $\lambda < 1$ is a constant that controls the exponential weight decay. Intuitively, the echoes that are created given shorter prefixes (e.g. the echo highlighted in Figure 1(b)) influence the final score more than